Group: Jason Antunes, [redacted]

Exploratory Data Analysis (EDA) Report

Descriptive Statistics & Distributions

Preliminary Overview

Initially, a high-level data profiling was conducted on the raw dataset (see the table that follows). This stage maintains the original business formats (HH:MM) to validate data completeness and identify initial boundaries, such as the maximum session duration of 04:58 and the peak charge of $1,012.50.

Columns 1-7

	Client	Subclient	Weekday	Date	Start	End	Total Time
Count	168	23	168	168	168	168	168
Average							01:06
Minimum				21-08-2025			00:02
Maximum				18-01-2026			04:58
Range				150			04:56

Columns 8-14

	Hourly Rate	Charge	Charge Notes	Invoice Date	Invoive Amount	USD	Notes
Count	168	168	48	71	12	12	139
Average	$117.63	$ 138.37			918.43		
Minimum	$ 40.00	$ 1.33		04-09-2025	132.75		
Maximum	$225.00	$1,012.50		29-11-2025	7,552.50		
Range	$185.00	$1,011.17		86	7,419.75		

Detailed Summary Statistics Table

For deeper quantitative analysis, we used Python to perform secondary data cleaning and standardization. By converting time into decimal hours and introducing metrics, such as standard deviation and quartiles, we established a robust analytical foundation. This step ensures precision in billing and provides a scientific baseline for forecasting and correlation analysis.

	Total Hours	Hourly Rate ($)	Charge Amount ($)
Count	168.00	168.00	168.00
Mean	1.1124	117.628	138.3671
Std Dev	0.863	57.7043	138.0169

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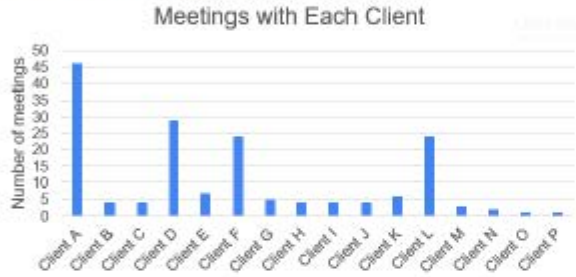
Minimum	0.0333	0	0
Median	0.9583	100	93.5833
Maximum	4.9667	225	1012.5

Visualizations and Insights

Number of Meetings per Client



Client A, D, F, and L occupy the most amount of meeting slots. Client A has 15 more meetings than the next highest client (Client D).

Days of the Week with the Most Meetings

Fairly even, with a small bump on Thursdays. Thursdays are the busiest day for meetings.

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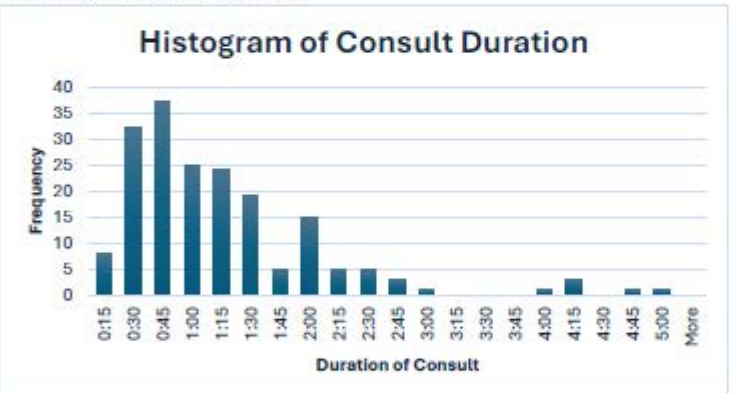
Frequency of Start Times for Meetings

Most meetings take place around lunchtime. Meetings also go late into the night.

Frequency of Consult Durations



Most are under an hour and a half with the most common running approximately around the 45 minutes mark. Odd times, they can go as long as 5 hours.

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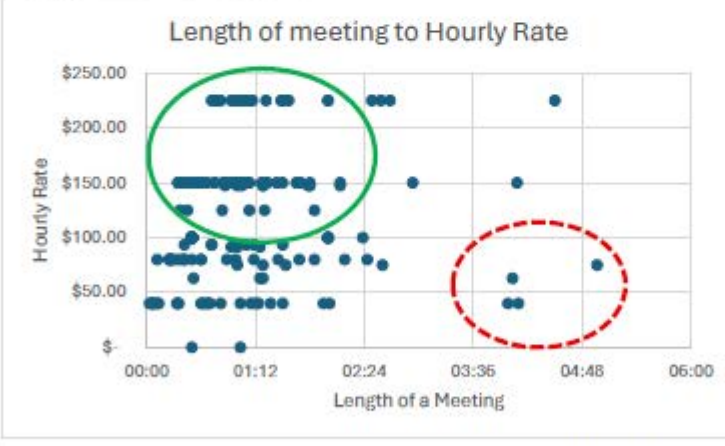
Key Visualizations and Insights

Consulting Time in a Day vs. Revenue

There is a fairly clear trend along the green line.

The points within the red circle show some resource allocation issues because there is a lot of time being spent on these projects, but low revenue.

Meeting Duration vs. Hourly Rate



We can see a couple pro bono consultations at the bottom of this chart in the bottom left corner of the chart.

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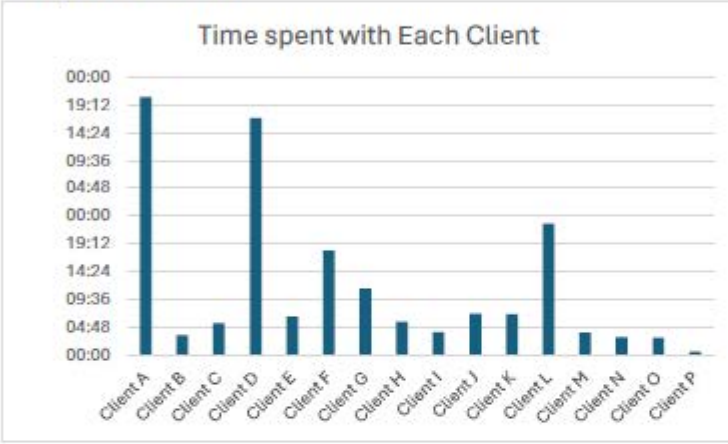
The solid green line surrounds the meetings and hourly rates that are very beneficial for the business. They are within the duration the histogram of duration showed us is most common and have high hourly rates to support revenue.

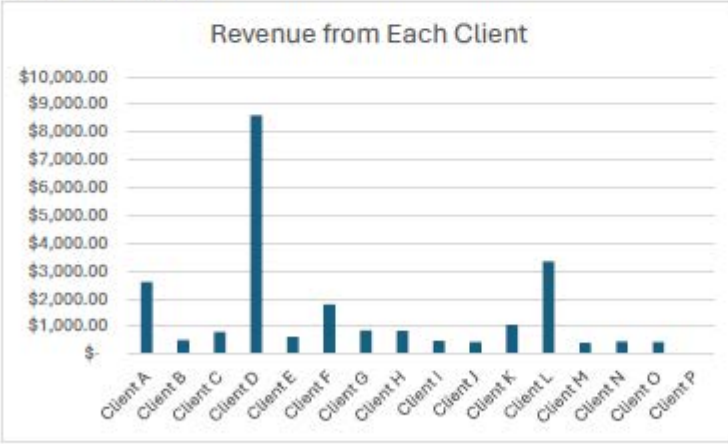
The dashed red circle indicates projects that are demanding lots of resources but should be re-evaluated for the pricing. A project that requires this amount of time is probably more complicated.

Breaking Down Consult Time vs. Revenue, and Meeting Duration vs. Hourly Rate

Time Spent with Each Client



Revenue from Each Client



These two charts above help identify what clients may be appearing in the two red circles above. Client A especially seems to have a high demand for time, but a low revenue.

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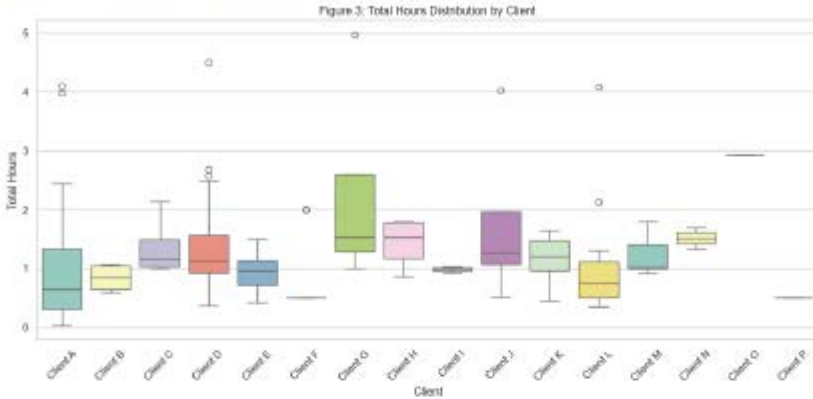
Distribution of Total Hours by Client



By comparing revenue (bar chart) with time investment (box plot), we found client D is our most valuable client, generating nearly $9,000 in revenue, but with fewer total hours worked than client A. In contrast, Client A consumes the most resources and has the most unpredictable workload (as shown by the outliers in the revenue chart) but provides very low compensation. This indicates the imbalance between input and output is the main operational risk we identified in the challenge statement.

Preliminary Findings & Hypotheses

Preliminary Findings

- **Market Concentration:** our data shows the consulting firm relies on a few frequent clients. While we have many clients, a small group (like Client A, D, and L) drives the majority of the total revenue and workload
- **The Efficiency Gap (Client D vs. A):** There is a clear mismatch between time spent and money earned. Client D is our most efficient client, generating about $9,000 with moderate hours. In contrast, Client A is like a "time sink," consuming the most total hours (about 24 hours) but receiving far less than return.
- **High Opportunity Cost:** we explored if spending less time on high-demand, low-compensation clients would clear up time for the several clients that are not as well established. After our analysis, it shows that the opportunity cost for Client A is high. Even if 20% of the time currently 'invested' in Client A's unpredictable long sessions were reallocated (see box plot outliers), the company could double the engagement of several smaller, less mature clients and then potentially discover new, high-value revenue streams similar to Client D.

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Hypotheses to Test

- **Hypothesis 1 (Pricing Optimization):** We hypothesize that raising the hourly rate for "high-demand" clients like Client A to meet the company average (approx. $117/hr) will fix the revenue-to-hour ratio and move them closer to the organizational trend line.
- **Hypothesis 2 (Resource Allocation):** If we reduce the time spent on Client A's unpredictable long sessions by 20% (based on the box plot outliers), we can double our engagement with smaller, newer clients and potentially find new high-value partners like Client D.
- **Hypothesis 3 (Time Policy):** According to the histogram of consultation duration, most meetings are under 1.5 hours but, in extreme cases, can reach 5 hours. Therefore, it is recommended to limit each meeting to 2 hours. This prevents a single client from taking up an entire day of an advisor's time and improves the team's overall operational efficiency.

Patterns, Anomalies & Outliers

Client	Subclient	Weekday	Date	Start	End	Total Time	Time (Mins) - Given	Hourly Rate	Charge - Calculated	Charge - given
Client A		Sun	12-10-2025	12:37	13:30	00:53	53	$ 80.00	$ 70.67	$1,168

Client	Subclient	Weekday	Date	Start	End	Total Time	Time (Mins) - Given
Client C		Thu	06-11-2025	13:00	14:00	01:00	73.75
Client C		Thu	06-11-2025	14:00	15:03	01:03	73.75
Client D		Tue	04-11-2025	13:00	13:56	00:56	62.5
Client D		Tue	04-11-2025	14:00	15:29	01:29	62.5

The highlighted data is from our source dataset – marked as Given in the column names. When total time or total charge was calculated through Excel, these discrepancies showed up.

There may be individual reasons why these are different from the calculations, or they may be due to human error. For the purposes of our work, we will be removing them and using the calculated amounts for data analysis.

Outliers

The below hourly rate is a one-time charge and is more than twice the next highest hourly rate. Deleting as an outlier. It was also an incomplete row, which is another reason to remove it.

Client	Subclient	Weekday	Date	Start	End	Total Time	Time (Mins)	Hourly Rate
Client F								$ 600.00

"Note: Programming and coordinating each session (idea, writing description, inviting guests, asking for bios)"

Possible Outliers Considered, but Retained

The minimum meeting duration is 2 minutes, which we worried would be an outlier. However there are 7 meetings that lasted less than 10 minutes. With a dataset this small we are hesitant to remove them from the analysis.